

MATH 490: ∞ -CATEGORY THEORY

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This survey was written as the summary for a directed reading course (MATH 490) at the University of Southern California in Spring 2024 under the guidance of Professor Joseph Helfer.

The theme of the course is ∞ -category theory. The course was conducted after a semester of algebraic topology course, so the basic notions of algebraic topology and category theory were assumed, and some more about them are introduced during the course. This survey aims to present a result in the end, so only the necessary contents are included, and some other contents are presented in the Appendix (if there is one).

These notes may contain mistakes. Some of the exercises and statements are not provided with a proof. Please use them with caution and verify the information as needed.

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1 Representable Functors

Yoneda Lemma

Definition 1.1. A category $\mathcal{C}$ is **locally small** if $\mathcal{C}(A, B)$ is a small set for any $A, B \in \mathcal{C}$.

Definition 1.2. Given a locally small category $\mathcal{C}$ and an object $A \in \mathcal{C}$, we have a functor:

$$\hat{A} : \mathcal{C}(-, A) : \mathcal{C}^{op} \rightarrow \mathbf{Set}.$$

A functor $F : \mathcal{C}^{op} \rightarrow \mathbf{Set}$ is **representable** if $F \cong \hat{A}$ for some $A \in \mathcal{C}$.

Lemma 1.3 (Yoneda Lemma). Let $\mathcal{C}$ be a locally small category. Let $F : \mathcal{C}^{op} \rightarrow \mathbf{Set}$ be a functor. Then for any $A \in \mathcal{C}$, the map

$$\mathbf{Set}^{\mathcal{C}^{op}}(\hat{A}, F) \rightarrow FA,$$

where a natural transformation $\alpha : \hat{A} \rightarrow F$ is mapped to $\alpha_A(1_A) \in FA$, is a bijection.

Proof. The natural transformation α is indicated below:

$$\begin{array}{ccc} \hat{A}A & \xrightarrow{\hat{A}f} & \hat{A}B \\ \downarrow \alpha_A & & \downarrow \alpha_B \\ FA & \xrightarrow{Ff} & FB \end{array}$$

where $f : B \rightarrow A$ is fixed.

Suppose that there is some other natural transformation β between $\hat{A}$ and F such that $\alpha_A(1_A) = \beta_A(1_A)$, then by commutativity of the square, we have

$$\alpha_B f = \alpha_B \hat{A}f(1_A) = Ff(\alpha_A(1_A)) = Ff(\beta_A(1_A)) = \beta_B \hat{A}f(1_A) = \beta_B f.$$

Therefore, $\alpha_B = \beta_B$ for any $B \in \mathcal{C}$ and $f : B \rightarrow A$. Therefore, $\alpha = \beta$, and our map is injective.

For surjectivity, we try to construct a natural transformation $\alpha : \hat{A} \rightarrow F$. Fix $x \in FA$. Let $\alpha_A(1_A) = x$. To make the diagram commute, we must have $\alpha_B \hat{A}f(1_A) = \alpha_B f = Ff(x)$. Therefore, we simply define $\alpha_B f = Ff(x)$ for any $f \in \hat{A}B = \mathcal{C}(B, A)$.

To show that α indeed defines a natural transformation, we want to verify that the diagram below commutes:

$$\begin{array}{ccc} \hat{A}B & \xrightarrow{\hat{A}t} & \hat{A}C \\ \downarrow \alpha_B & & \downarrow \alpha_C \\ FB & \xrightarrow{Ft} & FC \end{array}$$

where $t : C \rightarrow B$ is a morphism in $\mathcal{C}$. Let $f : B \rightarrow A$, then $\alpha_C \hat{A}t(f) = \alpha_C(ft) = Fft(x)$ and $Ft\alpha_B f = Ft(Ff(x)) = (Ft \circ Ff)(x) = Fft(x)$ as F is contravariant. Therefore, the diagram commutes for arbitrary $t : C \rightarrow B$, proving that α is indeed a natural transformation. $\square$

Corollary 1.4. There exists a functor $\mathfrak{y} : \mathcal{C} \rightarrow \mathbf{Set}^{\mathcal{C}^{op}}$, where $\mathfrak{y}A = \hat{A}$ and $f : A \rightarrow B$ is sent to $f_* : \hat{A} \rightarrow \hat{B}$, where for any $Y \in \mathcal{C}$ and $g : Y \rightarrow A$, we define $f_*(g) = f \circ g$.

The functor $\mathfrak{y}$ is called the **Yoneda embedding**. It is fully faithful, i.e., $\mathfrak{y} : \mathcal{C}(A, B) \rightarrow \mathbf{Set}^{\mathcal{C}^{op}}(\hat{A}, \hat{B})$ is a bijection.

Proof. This comes directly from Yoneda's lemma by setting $F = \hat{B}$. $\square$

Connection to Cohomology

Definition 1.5. Fix $n \geq 1$ and $G \in \mathbf{Ab}$, the **Eilenberg MacLane Space** is a topological space $K(G, n)$ with

$$\pi_k(K(G, n)) \cong \begin{cases} G & k = n \\ 1 & k \neq n \end{cases}$$

Theorem 1.6. The set of homotopy classes of pointed maps between an arbitrary topological space Y and $K(G, n)$, $[Y, K(G, n)]$, satisfies:

- (i) $[Y, K(G, n)]$ has a natural Abelian group structure.
- (ii) $[Y, K(G, n)] \cong H^n(Y; G)$.

A proof can be found in [May99].

Remark 1. In other words, in the category **HoTop** (formal definition appears later), where, naively speaking, can be viewed as **Top** quotiented by homotopy class of maps, the cohomology functor H^n is represented by $K(G, n)$.

Definition 1.7. Given a pointed space (X, x_0) , the **loop space** of (X, x_0) is defined by the set of loops of X at x_0 , denoted by $\Omega_{x_0} X$.

Notice that a loop at x_0 could be viewed as a continuous pointed map $(S^1, 0) \rightarrow (X, x_0)$, so we equip $\Omega_{x_0} X$ with compact-open topology.

Corollary 1.8. Since S^1 is compact Hausdorff, $f : Y \rightarrow \Omega_{x_0} X$ is continuous iff $f_* : Y \times S^1 \rightarrow X$ is continuous.

By first showing the case for $k = 1$ and then inducting on k by noticing that $\Omega^2(X, x_0) = \Omega_{c_{x_0}} \Omega_{x_0} X$, we have that

Proposition 1.9. $\pi_k(X) \cong \pi_{k-1}(\Omega X)$ for every (positive) integer k .

A proof can be found in [Hat02].

Definition 1.10. The **suspension** SX of a topological space X is $X \times I / \sim$, where $(x, 0) \sim (y, 0)$ and $(x, 1) \sim (y, 1)$ for any $x, y \in X$.

The **reduced suspension** ΣX is defined to be $SX/q(x_0 \times I)$.

By looking at the definitions, it is observed that

Proposition 1.11. For any pointed space X and Y , we have a bijection between $\mathbf{Top}(\Sigma X, Y)$ and $\mathbf{Top}(X, \Omega Y)$. In other words, Σ is the left adjoint of Ω .

Remark 2. The previous proposition does not seem to be directly related to the representability of the cohomology theory, but it turns out to be useful in the proof.

The main idea is that we have

$$\widehat{K(G, n)} \cong \widehat{K(G, n+1)} \circ \Sigma$$

as we have

$$\mathbf{HoTop}(-, K(G, n+1)) \cong \mathbf{HoTop}(-, \Omega K(G, n+1)) \cong \mathbf{HoTop}(-, K(G, n)),$$

where the last isomorphism holds because that $\pi_k(K(G, n)) \cong \pi_{k+1}(\Omega(K(G, n+1)))$ for every k and that we have the approximation theorem mentioned below.

Theorem 1.12 (CW Approximation Theorem). *For every topological space X , there is some CW complex Y and a map $f : Y \rightarrow X$ such that f is a weak homotopy equivalence.*

A proof can be found in [May99].

2 Simplicial Sets I

Basic Notion

Definition 2.1. The **simplex category** Δ is defined as the category with objects $[n] := \{0, 1, \dots, n\}$ for every non-negative integer n and morphisms $\sigma : [n] \rightarrow [m]$ that are order-preserving (i.e., $\sigma(i) \leq \sigma(j)$ if $i < j$).

Definition 2.2. Recall that a **semi-simplicial set** is defined as a functor $X : \Delta_{\text{inj}}^{\text{op}} \rightarrow \mathbf{Set}$, where $\Delta_{\text{inj}}^{\text{op}}$ is the subcategory of Δ such that all the morphisms are injective.

Now, a **simplicial set** is defined as a functor $X : \Delta^{\text{op}} \rightarrow \mathbf{Set}$.

Definition 2.3. Given $\alpha : [n-1] \rightarrow [n]$ injective with a unique $i \notin \text{im } \alpha$, we have $\alpha^* : X_n \rightarrow X_{n-1}$ if X is a simplicial set, and $\alpha^* \sigma$ is called the i -th **face** of $\sigma \in X_n$.

Similarly, given $\alpha : [n+1] \rightarrow [n]$ surjective with $\alpha(i) = \alpha(i+1)$, we have $\alpha^* : X_n \rightarrow X_{n+1}$, and $\alpha^* \sigma$ is called the i -th **degeneracy** of σ . $\alpha^* \sigma \in X_{n+1}$ is called a **degenerate**.

We denote the i -th face map as $\partial_{n,i}$ (or ∂_i) and the i -th degeneracy map as $s_{n,i}$ (or s_i).

Remark 3. There are some formal rules restricting the relations between the degeneracy maps and face maps, but the general idea is that up to a change of index (usually ± 1) we are able to commute the maps on the neighboring sets.

Definition 2.4. The **boundary simplicial set** $\partial \Delta^n$ is defined as the subfunctor of Δ^n such that $\partial \Delta_m^n$ is defined to be $\{\sigma : [n] \rightarrow [m], \sigma \text{ not surjective}\}$.

Definition 2.5. Given a functor $F : \mathcal{C} \rightarrow \mathbf{Set}$, $A : \mathcal{C} \rightarrow \mathbf{Set}$ is called a **subfunctor** of F if $A_X \subset F_X$ for every $X \in \mathcal{C}$, and for any $f : X \rightarrow Y$, we have $Ff(A_X) \subset A_Y$.

Corollary 2.6. $\partial \Delta^n$ is the sub-simplicial set generated by $\partial_0(1_{[n]}), \dots, \partial_n(1_{[n]})$, all of which elements in Δ_{n-1}^n . (i.e., the maps $[n-1] \rightarrow [n]$ such that i is not in the image).

Proof. Notice that $\partial \Delta^n$ is a sub-simplicial set containing A , where $A_{n-1} = \{\partial_j(1_{[n]})\}$ and $A_k = \emptyset$ for any other k .

Now, it suffices to show that for any Y sub-simplicial set of X containing A , we have $\partial \Delta_k^n \subset Y_k$ for every k .

For any $\sigma : [k] \rightarrow [n]$ not surjective, we can decompose σ as $\partial_j \circ f$ for some $f : [k] \rightarrow [n-1]$ and $\partial_j : [n-1] \rightarrow [n]$, if $j \notin \text{im } f$.

Notice that have $\sigma = \partial(f)(\partial_j)$, where $\partial_n(f) : \Delta_{n-1}^n \rightarrow \Delta_k^n$. Therefore, $\sigma \in Y_k$, as Y is a subfunctor, and $\partial_j \in Y_{n-1}$. $\square$

The Realization and Singular Simplicial Functor

Definition 2.7. Given a simplicial set X , the **geometric realization** is defined as

$$|X| = (\bigsqcup X_n \times \Delta^n) / \sim,$$

where Δ^n represents the standard n -simplex, and the equivalence relation is given explicitly by $(\sigma_1, \Delta^{n_1}) \sim (\sigma_2, \Delta^{n_2})$ iff there exists some (σ', Δ^k) and $\alpha_1, \alpha_2 : [k] \rightarrow [n_1], [n_2]$, such that $\alpha_1^* \sigma' = \sigma_1$ and $\alpha_2^* \sigma' = \sigma_2$.

In other words, $(\sigma_1, \Delta^{n_1}) \sim (\sigma_2, \Delta^{n_2})$ iff σ_1 and σ_2 are both the degenerates of some $\sigma' \in X_k$.

Proposition 2.8. $|-| : \mathbf{sSet} \rightarrow \mathbf{Top}$ is a functor.

Definition 2.9. The **singular functor** S is defined as the functor $\mathbf{Top} \rightarrow \mathbf{sSet}$ where $S_n X = \mathbf{Top}(\Delta^n \rightarrow X)$, where Δ^n represents the standard n -simplex in $\mathbf{Top}$.

As we have shown in the study of algebraic topology, we have

Corollary 2.10. *S is indeed a functor.*

Theorem 2.11. *For $X \in \mathbf{sSet}$ and $Y \in \mathbf{Top}$, we have*

$$\mathbf{Top}(|X|, Y) \cong \mathbf{sSet}(X, SY),$$

or in other words, $|-|$ is the left adjoint of S .

Proof. Notice that we have

$$\begin{aligned} \mathbf{Top}(|X|, Y) &\cong \bigsqcup \mathbf{Top}(|\Delta^n|, Y) \\ &\cong \bigsqcup S_n Y \\ &\cong \bigsqcup \mathbf{sSet}(\Delta^n, SY) \\ &\cong \mathbf{sSet}(X, SY) \end{aligned}$$

where the third isomorphism holds by applying Yoneda lemma. $\square$

Remark 4. *The idea of geometric realization and the singular functor is quite important and lead to the definition of nerves of categories, in some sense extending what we have done on topological spaces to categories.*

Kan Complex and Horn

Definition 2.12. *The sub-simplicial set (n, k) -**horn** is defined as $\Lambda_k^n \subset \partial\Delta^n \subset \Delta^n$ generated by $\partial_0(1_{[n]}), \dots, \partial_n(1_{[n]})$ except the k -th one.*

By looking at the definition, we have that

Remark 5. *If $m < n - 1$, then $(\Lambda_k^n)_m = \Delta_m^n$; if $m = n - 1$, then $(\Lambda_k^n)_m = \Delta_{n-1}^n \setminus \partial_k(1_{[n]})$; if $m > n - 1$, then $(\Lambda_k^n)_m$ are degenerates.*

The definition of the horn is mostly used for the following definition:

Definition 2.13. *A simplicial set S is a **Kan complex** if for every $n \geq 0$ and $0 \leq k \leq n$, the horn filling condition holds for S . In other words, for every $\sigma : \Lambda_k^n \rightarrow S$ in $\mathbf{sSet}$, there is a lift $\tilde{\sigma} : \Delta^n \rightarrow S$ such that the following diagram commutes:*

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\sigma} & S \\ i \downarrow & \nearrow \exists \tilde{\sigma} & \\ \Delta^n & & \end{array}$$

where i is the inclusion morphism.

In other words, the terminal morphism $S \rightarrow \Delta^0$ (the terminal object in $\mathbf{sSet}$) is a Kan fibration. The formal definition for Kan fibration (and related model categorical stuff) is included in the Appendix (or maybe not).

Remark 6. *Notice that the lift does not need to be unique in our definition.*

3 More Category Theory

Adjoint

Definition 3.1. Given functors $F : \mathcal{C} \rightarrow \mathcal{D}$, $G : \mathcal{D} \rightarrow \mathcal{C}$, F is said to be **left adjoint** to G if there exists a bijective map

$$\varphi_{X,Y} : \mathcal{D}(FX, Y) \rightarrow \mathcal{C}(X, GY)$$

for any $X, Y \in \mathcal{C}, \mathcal{D}$, such that $\varphi : \mathcal{D}(F-, -) \rightarrow \mathcal{C}(-, G-)$ between functors $\mathcal{C}^{op} \times \mathcal{D} \rightarrow \mathbf{Set}$ is a natural isomorphism.

φ is called an **adjunction**.

Definition 3.2. Given an adjunction φ between F and G , a **unit** is defined as a natural transformation $\eta : 1_{\mathcal{C}} \rightarrow GF$ defined by:

$$\eta_X : X \rightarrow GFX$$

where $\eta_X = \varphi_{X,FX}(1_{FX})$.

Similarly, a **counit** is defined as a natural transformation $\epsilon : FG \rightarrow 1_{\mathcal{D}}$ defined by:

$$\epsilon_Y : FG Y \rightarrow Y$$

where $\epsilon_Y = \varphi_{GY,Y}^{-1}(1_{GY})$.

Lemma 3.3. The map $FA \xrightarrow{f} B \xrightarrow{g} B'$ is in correspondence with $A \xrightarrow{\varphi_{A,B}(f)} GB \xrightarrow{Gg} GB'$.

Similarly, the map $A' \xrightarrow{f} A \xrightarrow{g} GB$ is in correspondence with $FA' \xrightarrow{Ff} FA \xrightarrow{\varphi_{A,B}^{-1}(g)} B$.

Proof. This comes from the fact that φ is a natural transformation. We therefore have the following diagram commute:

$$\begin{array}{ccc} \mathcal{D}(FA, B) & \xrightarrow{g_*} & \mathcal{D}(FA, B') \\ \downarrow \varphi_{A,B} & & \downarrow \varphi_{A,B'} \\ \mathcal{C}(A, GB) & \xrightarrow{Gg_*} & \mathcal{C}(A, GB') \end{array}$$

Therefore, for such a map $g \circ f$ comes from $g_*(f)$, and $\varphi_{A,B'}$ sends it to $Gg \circ \varphi_{A,B}(f)$.

Similarly, the latter claim is shown by considering the commutative diagram:

$$\begin{array}{ccc} \mathcal{D}(FA, B) & \xrightarrow{Ff_*} & \mathcal{D}(FA', B) \\ \downarrow \varphi_{A,B} & & \downarrow \varphi_{A',B} \\ \mathcal{C}(A, GB) & \xrightarrow{f_*} & \mathcal{C}(A', GB) \end{array}$$

□

By applying the above lemma to the following statement, we can easily prove that:

Theorem 3.4. The following diagrams commute, where φ is the adjunction between F and G , for every $X \in \mathcal{C}$ or $Y \in \mathcal{D}$:

$$\begin{array}{ccc} FX & \xrightarrow{F\eta_X} & FGFX \\ \searrow 1_{FX} & & \downarrow \epsilon_{FX} \\ & & FX \end{array} \quad \begin{array}{ccc} GY & \xrightarrow{\eta_{GY}} & GFGY \\ \searrow 1_{GY} & & \downarrow G\epsilon_Y \\ & & GY \end{array}$$

One result is that

Theorem 3.5. For any topological space X , SX is a Kan complex.

Proof. By the adjunction between $|-|$ and S , the diagram in **sSet**

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\sigma} & SX \\ i \downarrow & \nearrow \bar{\sigma} & \\ \Delta^n & & \end{array}$$

in **sSet** corresponds to the diagram

$$\begin{array}{ccc} |\Lambda_k^n| & \xrightarrow{\sigma} & X \\ i \downarrow & \nearrow \bar{\sigma} & \\ |\Delta^n| & & \end{array}$$

in **Top**. Notice that the topological (n, k) horn is a deformation retract of the standard n -simplex, so the lift for the second diagram exists. By adjunction we have the existence of lift for the first diagram. $\square$

More properties of adjoint functors will be (or not) included in the Appendix.

Examples of Adjoint

Example 1. As explained earlier, the loop space functor Ω is right adjoint to the suspension functor Σ . In other words,

$$\mathbf{Top}(\Sigma X, Y) \cong \mathbf{Top}(X, \Omega Y).$$

Example 2. As explained earlier, the realization functor $|-| : \mathbf{sSet} \rightarrow \mathbf{Top}$ is left adjoint to the singular simplicial functor $S : \mathbf{Top} \rightarrow \mathbf{sSet}$.

Example 3. Let $F : \mathbf{Set} \rightarrow \mathbf{Grp}$ be sending a set to the free group generated by it, and $U : \mathbf{Grp} \rightarrow \mathbf{Set}$ be the forgetful functor, then we have

$$\mathbf{Grp}(FX, G) \cong \mathbf{Set}(X, UG).$$

Definition 3.6. $\Delta : \mathcal{C} \rightarrow \mathcal{C} \times \mathcal{C}$ is the **diagonal** functor sending an object X to $X \times X$ and an arrow $f : X \rightarrow Y$ to $(f, f) : (X, X) \rightarrow (Y, Y)$.

Example 4. Let $\mathcal{C}$ be a category with finite product. Let $\times : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ be the functor sending (X, Y) to their Cartesian product $X \times Y$ in $\mathcal{C}$. We then have

$$\mathcal{C} \times \mathcal{C}(\Delta X, (Y, Z)) \cong \mathcal{C}(X, Y \times Z),$$

and it can be verified that the bijection is natural, so Δ is left adjoint to $\times$.

Exponent

Remark 7. In set, we typically use Y^X to denote the set of all functions from X to Y . Similarly, in **Top**, we use Y^X to denote the collection of all continuous functions from X to Y , which is a topological space itself.

Now, we hope to generalize this notion to whichever categories that are possible.

Definition 3.7. Given a category $\mathcal{C}$ that has finite products, let $X, Y \in \mathcal{C}$, the **exponent object** Y^X is defined by the following universal property: given any $f : A \times X \rightarrow Y$ in $\mathcal{C}$, there is a unique $\bar{f} : A \times X \rightarrow Y^X \times X$ such that the following diagram commutes:

$$\begin{array}{ccc} Y^X \times X & \xrightarrow{ev} & Y \\ \exists! \bar{f} \uparrow & \nearrow f & \\ A \times X & & \end{array}$$

where $ev : Y^X \times X \rightarrow Y$ is a fixed morphism called the **evaluation morphism**.

Remark 8. Clearly, the exponent object Y^X in set satisfies this condition, with the evaluation morphism given by $ev(f, x) = f(x)$.

Example 5. Notice that the condition is equivalent to the existence of the adjunction between the functors

$$(X \times -) : \mathcal{C} \rightarrow \mathcal{C}$$

and

$$(-)^X : \mathcal{C} \rightarrow \mathcal{C},$$

where the former is the left adjoint, as we have a bijection between $\mathcal{C}(X \times A, Y)$ and $\mathcal{C}(A, Y^X)$.

A result that we will use for the simplicial category is the following:

Theorem 3.8. For any small category $\mathcal{C}$, $\hat{\mathcal{C}} = \mathbf{Set}^{\mathcal{C}^{op}}$ has exponential objects.

Proof. (i) Fix $X, Y \in \hat{\mathcal{C}}$ and $A \in \mathcal{C}$, we want to have

$$\hat{\mathcal{C}}(\hat{A}, Y^X) \cong \hat{\mathcal{C}}(\hat{A} \times X, Y).$$

Notice that by Yoneda lemma, we have $\hat{\mathcal{C}}(\hat{A}, Y^X) \cong Y^X(A)$, so we can therefore define $Y^X \in \hat{\mathcal{C}}$ by sending $A \in \mathcal{C}$ to $\hat{\mathcal{C}}(\hat{A} \times X, Y)$.

- (ii) We first show that the defined $Y^X : \mathcal{C}^{op} \rightarrow \mathbf{Set}$ is indeed a functor. Given $f : A \rightarrow B$, $g : B \rightarrow C$ morphisms in $\mathcal{C}$, we have the induced morphism $\hat{f} : \hat{A} \rightarrow \hat{B}$ and $\hat{g} : \hat{B} \rightarrow \hat{C}$ in $\hat{\mathcal{C}}$; also, $\hat{g} \circ \hat{f} = \widehat{gf}$. Therefore, we have $Y^X(gf) : Y^X(C) \rightarrow Y^X(A) = Y^X(f) \circ Y^X(g)$. Also $1_A : A \rightarrow A$ induces $1_{\hat{A}}$, which is clearly sent to $Y^X(1_A) = 1_{Y^X(A)}$ by our definition. Therefore, Y^X is a functor.
- (iii) Now, we define the evaluation morphism $ev : Y^X \times X \rightarrow Y$ by $ev(T, x_A) = T(1_A, x_A)$, where $A \in \mathcal{C}$, $T \in \hat{\mathcal{C}}(\hat{A} \times X, Y)$, $1_A \in \mathcal{C}(A, A) = \hat{A}(A)$, and $x_A \in X(A)$.

We now show that this is indeed a natural transformation, which is equivalent to show that the following diagram commute for any $A, B \in \mathcal{C}$ with $f : B \rightarrow A$:

$$\begin{array}{ccc} Y^X(A) \times X(A) & \xrightarrow{\hat{f} \times Xf} & Y^X(B) \times X(B) \\ \downarrow ev_A & & \downarrow ev_B \\ Y(A) & \xrightarrow{Yf} & Y(B) \end{array}$$

Let $T \in \hat{\mathcal{C}}(\hat{A} \times X, Y)$ and $x_A \in X(A)$. (T, x_A) is mapped to $T(1_A, x_A) \in Y(A)$ and to $(T \circ (\hat{f} \times 1_X), Xf(x_A)) \in Y^X(B) \times X(B)$.

By our definition, $ev_B((T \circ (\hat{f} \times 1_X), Xf(x_A))) = (T \circ (\hat{f} \times 1_X))(1_B, Xf(x_A)) = T(f, Xf(x_A))$, and we need to show that this equals $Yf(T(1_A, x_A))$.

Notice that $T \in \hat{\mathcal{C}}(\hat{A} \times X, Y)$, so it is itself a natural transformation, and we therefore have the following commutative diagram:

$$\begin{array}{ccc} \hat{A}(A) \times X(A) & \xrightarrow{\hat{f} \times Xf} & \hat{A}(B) \times X(B) \\ \downarrow T_A & & \downarrow T_B \\ Y(A) & \xrightarrow{Yf} & Y(B) \end{array}$$

Plugging $(1_A, x_A)$ into the top-left of the square, we have that $Yf(T(1_A, x_A)) = T(f, Xf(x_A))$, which is exactly what we desire. Therefore, ev is indeed a natural transformation.

- (iv) Last but not least, we have to show that Y^X has the needed universal property.

Now, we need to construct $\bar{F} : Z \rightarrow Y^X$ given a functor $F : Z \times X \rightarrow Y$. Notice that this is equivalent to have a map $Z(A) \rightarrow \hat{\mathcal{C}}(\hat{A} \times X, Y)$ by the definition of Y^X .

Now, let $z_A \in Z(A)$ be fixed. We have an induced map $F|_{z_A} : X(A) \rightarrow Y(A)$, which would be our map $\hat{F}_{z_A} : \hat{A}(A) \times X(A) \rightarrow Y(A)$ when the first coordinate being 1_A . In other words, $\hat{F}_{z_A}(1_A, x_A) = F|_{z_A}(x_A)$ for every $x_A \in X(A)$.

Notice that by Yoneda lemma, $Z(A) \cong \hat{\mathcal{C}}(\hat{A}, Z)$ so z_A corresponds to some natural transformation $T_{z_A} : \hat{A} \rightarrow Z$.

Now, for an arbitrary $f : B \rightarrow A$, we must have a map $T_{z_A, B} : \hat{A}(B) \rightarrow Z(B)$, which sends f to $z_B = T_{z_A, B}(f) \in Z(B)$. Notice that z_B defines a map $F|_{z_B} : X(B) \rightarrow Y(B)$ by the same reasoning as for z_A , and we will simply define $\hat{F}_{z_A}(f, x_B) = F|_{z_B}(x_B)$.

The diagram below clearly commutes by our definition, and is clearly unique because for every z_A we have only this way of defining it so that the diagram commute with respect to our definition of the evaluation morphism. The last step is to verify that $\hat{F}_{z_A}$ is indeed a natural transformation.

It is equivalent to verify that the following diagram commute:

$$\begin{array}{ccc} \hat{A}(B) \times X(B) & \xrightarrow{\hat{g} \times Xg} & \hat{A}(C) \times X(C) \\ \downarrow \hat{F}|_{T_{z_A, B}} & & \downarrow \hat{F}|_{T_{z_A, C}} \\ Y(B) & \xrightarrow{Yg} & Y(C) \end{array}$$

where $z_A \in Z(A)$ is fixed, $g : C \rightarrow B$ is a morphism in $\mathcal{C}$. Fix (f, x_B) on the top-left corner of the diagram. It is sent to $\hat{F}|_{T_{z_A, B}(f)}(x_B) = F(T_{z_A, B}(f), x_B)$ in $Y(B)$ and to $(gf, Xg(x_B))$ in $\hat{A}(C) \times X(C)$.

Therefore, we have to show that $Y_g(F(T_{z_A, B}(f), x_B)) = F(T_{z_A, C}(gf), Xg(x_B))$ in $Y(C)$. Let $T_{z_A, B}(f) = z_B$, then $T_{z_A, C}(gf) = Zg(z_B)$. Therefore, we would need to show that $Y_g(F(z_B, x_B)) = F(Zg(z_B), Xg(x_B))$, which holds exactly because F is a natural transformation, so the following diagram commute:

$$\begin{array}{ccc} Z(B) \times X(B) & \xrightarrow{Zg \times Xg} & Z(C) \times X(C) \\ \downarrow F_B & & \downarrow F_C \\ Y(B) & \xrightarrow{Yg} & Y(C) \end{array}$$

and the equality comes by plugging in (z_B, x_B) on the top-left corner.

Therefore, the universal property is met, and Y^X is indeed the exponential object in $\hat{\mathcal{C}}$.

□

Definition 3.9. A category $\mathcal{C}$ is **Cartesian closed** if it has exponential objects and finite products.

Localization

The homotopy category (**HoTop**, for example), has a fuzzy definition. Now, we try to formalize the definition by introducing the concept of the localization of a category.

Definition 3.10. Let $\mathcal{C}$ be a category and $W \subset \text{Ar}\mathcal{C}$. The **localization** $\mathcal{C}[W^{-1}]$ is defined by the following rules:

- (i) The objects of $\mathcal{C}[W^{-1}]$ are exactly $\text{Ob}\mathcal{C}$.
- (ii) The morphisms are the set of strings

$$\{a_0 \xrightarrow{f_1^{\epsilon_1}} a_1 \xrightarrow{f_2^{\epsilon_2}} \dots \xrightarrow{f_n^{\epsilon_n}} a_n, \epsilon_j = \pm 1, f_j \in \left\{ \begin{array}{ll} \mathcal{C}(a_{j-1}, a_j) \cap W & \epsilon_j = 1 \\ \mathcal{C}(a_j, a_{j-1}) \cap W & \epsilon_j = -1 \end{array} \right\}\}$$

modulo the equivalence relation:

- $a_0 \xrightarrow{f} a_1 \xrightarrow{g} a_2 \sim a_0 \xrightarrow{g \circ f} a_2$.
- If $a_0 \xrightarrow{f} a_1$ is an isomorphism, then $a_0 \xrightarrow{f} a_1 \xrightarrow{f^{-1}} a_0 \sim a_0 \xrightarrow{1_{a_0}} a_0$ and $a_1 \xrightarrow{f^{-1}} a_0 \xrightarrow{f} a_1 \sim a_1 \xrightarrow{1_{a_1}} a_1$.

- The empty string is equivalent to $a \xrightarrow{1_a} a$ for every $a \in \mathcal{C}$.
- If $a_1 \xrightarrow{f} a_2 \sim a_1 \xrightarrow{g} a_2$ then $a_1 \xrightarrow{f} a_2 \xrightarrow{h} a_3 \sim a_1 \xrightarrow{g} a_2 \xrightarrow{h} a_3$ and $a_0 \xrightarrow{h} a_1 \xrightarrow{f} a_2 \sim a_0 \xrightarrow{h} a_1 \xrightarrow{g} a_2$ for every $a_0 \xrightarrow{h} a_1$ and $a_2 \xrightarrow{h} a_3$.
- The source and target of each arrow is the first and the last object in the string.

Remark 9. In general, this construction is very similar to the construction of the free group: both of which treat the object in the group or the arrows in the category as a set of strings modulo some equivalence relation, which in some sense respect the original operation in the given group or category.

Similar to the free group, the construction above is not that important, and the accompanying universal property is more crucial. In fact, there are two versions of the universal property.

The more straightforward (but less useful) one is:

Theorem 3.11. If $\gamma : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ and $F : \mathcal{C} \rightarrow \mathcal{D}$ are functors such that γw and Fw are isomorphisms in $\mathcal{C}[W^{-1}]$ and $\mathcal{D}$ for every $w \in W$, then we have a unique functor $\bar{F} : \mathcal{C}[W^{-1}] \rightarrow \mathcal{D}$ such that the following diagram commute:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\gamma} & \mathcal{C}[W^{-1}] \\ & \searrow F & \downarrow \bar{F} \\ & & \mathcal{D} \end{array}$$

This property determines $\mathcal{C}[W^{-1}]$ up to an isomorphism.

On the other hand, there is a weaker version:

Theorem 3.12. If $\gamma : \mathcal{C} \rightarrow \mathcal{C}[W^{-1}]$ and $F : \mathcal{C} \rightarrow \mathcal{D}$ are functors such that γw and Fw are isomorphisms in $\mathcal{C}[W^{-1}]$ and $\mathcal{D}$ for every $w \in W$, then we have a unique functor $\bar{F} : \mathcal{C}[W^{-1}] \rightarrow \mathcal{D}$ such that the following diagram commute up to equivalence:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{\gamma} & \mathcal{C}[W^{-1}] \\ & \searrow F & \downarrow \bar{F} \\ & & \mathcal{D} \end{array}$$

This property determines $\mathcal{C}[W^{-1}]$ up to an equivalence.

A proof can be found in [KS06]

Remark 10. This universal property is very similar to the one for the localization of a ring: sending every $w \in W$ to an isomorphism is the same as sending an element in a ring A to some unit in another ring B , which induces a unique lift from the localization $S^{-1}A$ to B .

Remark 11. A famous issue that arises is that $\mathcal{C}[W^{-1}]$ might not be small even if $\mathcal{C}$ is small. However, if $\mathcal{C}$ is a model category and W is the weak equivalences (definition in the Appendix), then $\mathcal{C}[W^{-1}]$ is small.

Homotopy Category

An important result from the definition of localization (and some model theory) is that

Theorem 3.13. $\mathbf{HoTop} \simeq \mathbf{HosSet}$.

A proof for this can be found in [May99].

Remark 12. In the above theorem, the homotopy category refers to the localization over the weak equivalences $\mathcal{W}_{\mathbf{Top}}$ and $\mathcal{W}_{\mathbf{sSet}}$ in the two categories. $\mathcal{W}_{\mathbf{Top}}$ is the set of weak homotopy equivalences between topological spaces, while $\mathcal{W}_{\mathbf{sSet}}$ refers to morphisms f between simplicial set such that $|f| \in \mathcal{W}_{\mathbf{sSet}}$.

Remark 13. *The proof of this theorem is more involved and requires theories about model structures on **Top** and **sSet**.*

It is also noticeable that the theorem is simply a special case of the Quillen equivalences between categories, induced by the Quillen adjunction between functors.

In particular, $|-|$ and S are not only adjoint but also form a Quillen adjunction pair. Of course, showing that they form such a pair is the main difficulty in the proof.

In fact, there is a further equivalence relation between **HoKan** and **HoCW**, the homotopy category of Kan complexes and CW complexes.

Recall that a weak homotopy equivalence induces isomorphisms on homotopy groups π_n for every n . Additionally, every topological space is weak equivalent to some CW complex according to **1.12**.

Therefore, by mapping each space to the CW complex that it is weak equivalent to gives:

Corollary 3.14. **HoTop** $\simeq$ **HoCW**.

Moreover, assuming the foundation of model theory (the existence of Quillen's model structure on **sSet**, which can be found in [Qui67]), we have that

Corollary 3.15. **HosSet** $\simeq$ **HoKan**.

Proof. For every simplicial set X , there is the unique morphism $U_X : X \rightarrow \Delta^0$. Now, by Quillen's model structure on **sSet**, there exists a decomposition of U_X into a weak equivalence followed by a fibration, in the form below:

$$\begin{array}{ccc} & Y & \\ W \nearrow & & \searrow U_Y \\ X & \xrightarrow{U_X} & \Delta^0 \end{array}$$

where U_Y is the unique morphism from Y to Δ^0 .

Now, since U_Y has to be a fibration, Y must be a fibrant object, or in other word, a Kan complex. Since W is a weak equivalence between X and Y , we conclude that X , and therefore any simplicial set, is weak equivalent to some Kan complex, leading to the equivalence between the two categories. $\square$

All in all, the four categories below are all equivalent:

Corollary 3.16. **HoKan** $\simeq$ **HosSet** $\simeq$ **HoTop** $\simeq$ **HoCW**.

4 Simplicial Sets II

Quasi-Category and Inner Horn

Another concept that we are going to use frequently is the following:

Definition 4.1. A simplicial set S is a **quasi-category** if for every $n \geq 0$ and $0 < k < n$, the inner horn filling condition holds for S . In other words, for every $\sigma : \Lambda_k^n \rightarrow S$ in $\mathbf{sSet}$, there is a lift $\tilde{\sigma} : \Delta^n \rightarrow S$ such that the following diagram commutes:

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\sigma} & S \\ \downarrow i & \nearrow \exists \tilde{\sigma} & \\ \Delta^n & & \end{array}$$

where i is the inclusion morphism.

Remark 14. The difference between a quasi-category and a Kan complex is that S does not need have lifts for outer horns (i.e., for $\sigma : \Lambda_0^n \rightarrow S$ or $\sigma : \Lambda_n^n \rightarrow S$).

It is clear that a Kan complex is always a quasi-category.

Nerve

Similar to how we define a functor from $\mathbf{Top}$ to $\mathbf{sSet}$, we do the same for the category of categories, $\mathbf{Cat}$.

Definition 4.2. We define a category $[n]$ for each $n \geq 0$, where the objects are $\{0, 1, \dots, n\}$ and there is a unique morphism from i to j whenever $i \leq j$.

Now, we define the **nerve** of a category $\mathcal{C}$ to be $NC \in \mathbf{sSet}$, where $(NC)_k = \mathbf{Cat}([k], \mathcal{C}) = \{a_0 \xrightarrow{f_1} a_1 \xrightarrow{f_2} \dots \xrightarrow{f_k} a_k, a_j \in \mathcal{C}, 0 \leq j \leq k\}$.

To make NC into a simplicial set, we need to define the face and degeneracy maps. Here, given a composable morphism $F := a_0 \xrightarrow{f_1} a_1 \xrightarrow{f_2} \dots \xrightarrow{f_k} a_k$ in $(NC)_k$, the i -th face is

$$\partial_i^k F = a_0 \xrightarrow{f_1} a_1 \xrightarrow{f_2} \dots \xrightarrow{f_i} a_i \xrightarrow{f_{i+1} \circ f_i} a_{i+1} \xrightarrow{f_{i+2}} \dots \xrightarrow{f_k} a_k$$

if $0 < i < k$. If $i = 0$ (k), we simply remove the first (last) object and morphism in F .

On the other hand, the i -th degeneracy map is defined by

$$s_i^k F = a_0 \xrightarrow{f_1} a_1 \xrightarrow{f_2} \dots \xrightarrow{f_{i-1}} a_{i-1} \xrightarrow{f_i} a_i \xrightarrow{1_{a_i}} a_i \xrightarrow{f_{i+1}} \dots \xrightarrow{f_k} a_k,$$

i.e., by inserting an extra identity morphism at a_i .

It is not difficult to verify that NC is indeed a simplicial set and that $N : \mathbf{Cat} \rightarrow \mathbf{sSet}$ is a functor.

In fact, we have

Theorem 4.3. A quasi-category S satisfies the unique inner horn filling condition if and only if $S \simeq NC$ for some category $\mathcal{C}$.

In other words, if for every $\sigma : \Lambda_k^n \rightarrow S$, $n \geq 0$, $0 < k < n$, there exists a unique $\bar{\sigma} : \Delta^n \rightarrow S$ such that the following diagram commutes:

$$\begin{array}{ccc} \Lambda_k^n & \xrightarrow{\sigma} & S \\ \downarrow i & \nearrow \exists! \bar{\sigma} & \\ \Delta^n & & \end{array}$$

then $S \simeq NC$.

A proof of this can be found in [Lur09].

Let $\mathbf{qCat}$ be the subcategory of quasi-categories of simplicial sets, we have that

Corollary 4.4. *Restricting the codomain of N , we have a functor $N : \mathbf{Cat} \rightarrow \mathbf{qCat}$.*

Remark 15. *While it is difficult to prove the above theorem, we will be giving an intuition for why the theorem holds when $n = 2$.*

A morphism $\sigma : \Lambda_1^2 \rightarrow NC$ can be represented by:

$$\begin{array}{ccc} & x_1 & \\ f \nearrow & & \searrow g \\ x_0 & & x_2 \end{array}$$

where x_0, x_1 , and x_2 are objects in $\mathcal{C}$ (therefore 0-simplices in NC), and f and g are morphisms in $\mathcal{C}$ (therefore 1-simplices in NC).

This horn can indeed be filled as we have the (unique) composition $g \circ f : x_0 \rightarrow x_2$, so that we can fill the horn by $\bar{\sigma} : \Delta^2 \rightarrow NC$, as shown in the diagram below:

$$\begin{array}{ccc} & x_1 & \\ f \nearrow & & \searrow g \\ x_0 & \xrightarrow{g \circ f} & x_2 \end{array}$$

Notice that the uniqueness comes exactly from the fact that compositions in a category is unique.

On the other hand NC fails to be a Kan complex, as if we have a morphism $\sigma : \Lambda_0^2 \rightarrow NC$

$$\begin{array}{ccc} & x_1 & \\ & \searrow g & \\ x_0 & \xrightarrow{h} & x_2 \end{array}$$

then it is not clear if there exists some $f : x_0 \rightarrow x_1$ such that $h = g \circ f$.

In fact, NC is a Kan complex (with unique fillings) if and only if $\mathcal{C}$ is a groupoid.

Composition and Homotopy

We now wish to find some functor that sends every quasi-category to a category. It seems natural to define the 0-simplices in a quasi-category to be the objects and the 1-simplices to be the morphisms, where the source and the target is the boundary 0-simplices. A difficulty arises is the definition of the composition of two morphisms: as we have discussed, a morphism $\sigma : \Lambda_1^2 \rightarrow S$ may not have a unique filling, so the composition of two morphisms may not be defined uniquely. In contrast, there could be a number of possibilities for the composition, and while they are not equal, we demand them to be homotopic.

Definition 4.5. A **homotopy** between $f, g : x \rightarrow y$ in a quasi-category S refers to a 2-simplex $\mathcal{K}$ such that $d_0\mathcal{K} = f$, $d_1\mathcal{K} = g$, and $d_2\mathcal{K} = 1_x$. In other words, the following diagram forms the boundary of $\mathcal{K}$:

$$\begin{array}{ccc} & x & \\ 1_x \nearrow & & \searrow f \\ x & \xrightarrow{g} & y \end{array}$$

Some definition requires f to be on the 2nd face and 1_y to be on the 0-face. It could be shown that the two definitions are equivalent by filling to a 3-horn.

Theorem 4.6. *The above definition of homotopy gives an equivalence relation on a quasi-category S , and it therefore makes sense to define the (in the naive sense) homotopy category of S , $\mathbf{Ho}S$, where the objects are the 0-simplices and the morphisms are the homotopy classes of 1-simplices.*

Proof. Given $f : x \rightarrow y$, let $\mathcal{K}_f = s_0 f : \Delta^2 \rightarrow S$. We then have $d_2\mathcal{K}_f = d_2 s_0 f = s_0 d_1 f = 1_x$. On the other hand, $d_0\mathcal{K}_f = d_1\mathcal{K}_f = f$, so $f \sim f$. Therefore, homotopy is reflexive.

Given another 1-simplex $g : x \rightarrow y$ such that $\sigma \in S_2$ serves as a homotopy between f and g , we construct an inner horn: $(\sigma, \mathcal{K}_g, \cdot, \mathcal{K}_{1_x}) : \Lambda_2^3 \rightarrow S$, where $\cdot$ refers to a face missing. By filling it to a 3-simplex: $\Delta^3 \rightarrow S$

and working it out geometrically, we conclude that the face $\cdot$ is infact a homotopy from g to f . Therefore, homotopy is symmetric.

The proof for transitivity is not included here. (I believe it could be done in the same way: construct a 3-horn and extend it to a 3-simplex such that the new face is a homotopy between f and h ; it is quite clear by drawing on paper.) $\square$

We now define compositions on a quasi category.

Definition 4.7. Given $f : x \rightarrow y$, $g : y \rightarrow z$ in a quasi-category S , a 1-simplex $h : x \rightarrow z$ is a **composition** for f and g if there exists some 2-simplex σ such that $d_0\sigma = g$, $d_1\sigma = h$, and $d_2\sigma = f$.

The following diagram is often used to denote such a composition:

$$\begin{array}{ccc} & y & \\ f \nearrow & \Downarrow \sigma & \searrow g \\ x & \xrightarrow{h} & z \end{array}$$

Theorem 4.8. Compositions of two 1-simplices in a quasi category are equal up to a homotopy.

A proof for this can be found in [Vai08].

Therefore, $\mathbf{Ho}S$ can be seen as an ordinary category, as the compositions are now unique and respect the equivalence relation of homotopy.

We therefore can define:

Definition 4.9. $\mathbf{Ho} : \mathbf{qCat} \rightarrow \mathbf{Cat}$ is the functor sending a quasi-category S to its homotopy category $\mathbf{Ho}S$.

Theorem 4.10. $\mathbf{Ho}$ is the left adjoint of N .

A proof for this can be found in [Lur09].

In fact, the nerve functor is fully faithful, so we have the following equivalence induced by the counit of the adjunction:

Corollary 4.11. For any category $\mathcal{C}$, $\mathbf{Ho}N\mathcal{C} \simeq \mathcal{C}$.

5 ∞ -Category Theory

Enriched Category

An important concept that we have used throughout category theory is the hom-set, where $\mathcal{C}(X, Y)$ denotes the set of morphisms from X to Y .

Now, we try to put more structure on categories by introducing the concept of enriched categories, which allow us to define ∞ -categories, as shown in this chapter.

Definition 5.1. *Given a category $\mathcal{V}$ with finite product and a terminal object, a $\mathcal{V}$ -enriched category $\mathcal{C}$ contains:*

- (i) A set $\text{Ob}\mathcal{C}$.
- (ii) For every $A, B \in \mathcal{C}$, $\mathcal{C}(A, B) \in \mathcal{V}$.
- (iii) A composition morphism $\mathcal{C}(A, B) \times \mathcal{C}(B, C) \rightarrow \mathcal{C}(A, C)$ in $\mathcal{V}$, denoted by $\circ$.
- (iv) An identity morphism $\mathbf{1} \rightarrow \mathcal{C}(A, A)$ in $\mathcal{V}$ for every $A \in \mathcal{C}$, where $\mathbf{1}$ is the terminal object in $\mathcal{V}$.
- (v) Associativity. The following diagram commutes:

$$\begin{array}{ccc} \mathcal{C}(A, B) \times \mathcal{C}(B, C) \times \mathcal{C}(C, D) & \xrightarrow{\circ \times 1} & \mathcal{C}(A, C) \times \mathcal{C}(C, D) \\ \downarrow 1 \times \circ & & \downarrow \circ \\ \mathcal{C}(A, B) \times \mathcal{C}(B, D) & \xrightarrow{\circ} & \mathcal{C}(A, B) \times \mathcal{C}(A, D) \end{array}$$

Example 6. A famous example would be **Ab**-enriched categories, where $\mathcal{C}$ is called an Abelian category if $\mathcal{C}(A, B)$ is an Abelian group and the compositions are Abelian group homomorphisms.

Remark 16. For any $\mathcal{V}$ with finite product and a terminal object, we could define a category $\mathcal{V} - \mathbf{Cat}$, containing all $\mathcal{V}$ -enriched categories. The functors are defined as usual, except that every map on the hom-sets are now morphisms in $\mathcal{V}$.

Definition 5.2. A **strict 2-category** is a **Cat**-enriched category.

By our definition earlier, composition and identity are now defined as functors. Associativity is defined by strict equality. Explicitly if

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

is a sequence of morphisms in a 2-category $\mathcal{C}$, then $(hg)f = h(gf)$ in $\mathcal{C}(A, D)$.

Additionally, for $f, g \in \mathcal{C}(X, Y)$, there would exist a hom-set between f and g , namely $\mathcal{C}(X, Y)(f, g)$, which is a set. The elements of these sets are often called the **2-morphisms** of $\mathcal{C}$. Clearly, the 2-morphisms lying in any $\mathcal{C}(X, Y)$ should satisfy the conditions we have for a 1-category, since $\mathcal{C}(X, Y)$ is itself a category.

Remark 17. We could view a 2-category as a 2-graph with some additional constraints. A 2-graph is, explicitly, the following sequence of maps between sets:

$$X_2 \xrightleftharpoons[t]{s} X_1 \xrightleftharpoons[t]{s} X_0$$

such that $s \circ s = s \circ t$ and $t \circ s = t \circ t$.

More explicitly, X_0 represents the objects in $\mathcal{C}$, X_1 represents the morphisms in $\mathcal{C}$, and X_2 represents the 2-morphisms in the enriched categorical structure of $\mathcal{C}$. Additionally, t and s are the maps sending a morphism to its target and source. The additional structure on the 2-graph would assure that the compositions work as defined.

Proposition 5.3. *There exists a 1-category, **r-Cat**, containing all the r -categories for any $r \geq 1$. It has finite product.*

We will not bother prove this, but it should be cautioned that the morphisms (or functors) in $\mathbf{rCat}$ should respect not only the compositions on 1-morphisms but also on higher morphisms.

A proof for the above proposition can be found in [Lur09].

Definition 5.4. By induction, we define an n -category to be an **(n-1)-Cat-enriched category**.

In other words, $\mathcal{C}(X, Y)$ is a $n - 1$ -category for every $X, Y \in \mathcal{C}$. There therefore are $(n - 1)$ -morphisms in $\mathcal{C}(X, Y)$, which become the n -morphisms in $\mathcal{C}$.

Definition 5.5. A ω -category is an object of $\lim_{n \rightarrow \infty} \mathbf{n-Cat}$. Concretely, a ω -category $\mathcal{C}$ can be viewed as a sequence $\{\mathcal{C}_n\}$ where each $\mathcal{C}_n$ is a n -category, and $\mathcal{C}_{n-1}$ is the underlying $(n - 1)$ -category of $\mathcal{C}_n$.

Remark 18. What we meant for the underlying $(n - 1)$ -category is the $(n - 1)$ -category formed by forgetting the n -morphisms of $\mathcal{C}_n$.

For example, every 1-category $\mathcal{C}$ can be viewed as a set by forgetting the morphisms, and every 2-category can be viewed as a 1-category by forgetting the morphisms in its category of morphisms (i.e., the 2-morphisms).

To be more clear, if we view $\mathcal{C}_n$ as a n -graph

$$X_n \xrightarrow[r]{r} \dots \xrightarrow[r]{r} X_1 \xrightarrow[r]{r} X_0$$

then its underlying $n - 1$ -category is simply by removing the leftmost set and arrows. We can therefore view a ω -category as a graph with infinite length.

Remark 19. We have defined 2-categories in a way such that for

$$A \xrightarrow{f} B \xrightarrow{g} C \xrightarrow{h} D$$

in a 2-category $\mathcal{C}$, we must have $(hg)f = h(gf)$ in $\mathcal{C}(A, D)$. However, since $\mathcal{C}(A, D)$ is a category itself, it seems more natural to ask for the existence of some isomorphism $\alpha_{f,g,h} : (hg)f \cong h(gf)$ in $\mathcal{C}(A, D)$.

Definition 5.6. A **bicategory** is defined in the same way as a 2-category except that associativity asks for an isomorphism instead of a strict equality. A bicategory is also called a **weak 2-category**.

Example 7. An example for a bicategory is $\mathcal{C}$ with $\text{Ob}\mathcal{C} = \mathbf{Rng}$ and $\mathcal{C}(R, S) = \mathbf{RMod}_S$, the Abelian groups that are simultaneously R -left module and S -right module.

Composition is defined by $(-) \otimes_S (-) : \mathcal{C}(R, S) \times \mathcal{C}(S, T) \rightarrow \mathcal{C}(R, T)$, and the property of tensor product gives that

$$(L \otimes_S M) \otimes_T N \cong L \otimes_S (M \otimes_T N),$$

so $\mathcal{C}$ is a bicategory but fails to be a strict 2-category.

Similarly one could define a tricategory and even higher weaker categories, which would be tedious to define explicitly. Some more ideas about weak categories and their relationship with monoidal categories are included in the Appendix.

Homotopy Hypothesis

We now try to extend the concept of a groupoid.

Definition 5.7. A **strict (n,k)-category** is an n -category such that all cells of dimension $k + 1$ or above are isomorphisms.

A $(n, 0)$ -category is called a **n-groupoid**.

Corollary 5.8. Let $\mathcal{C}$ be a (n, k) -category, then for any $A, B \in \mathcal{C}$, $\mathcal{C}(A, B)$ is a $(n - 1, k - 1)$ -category.

Definition 5.9. *There is a $(\infty, 1)$ -category, called the category of **spaces**, denoted as **Spc**.*

The objects are topological spaces, the 1-morphisms are continuous maps between topological spaces, the 2-morphisms are homotopies, the 3-morphisms are homotopies between homotopies, etc.

We are going to see another perspective for **Spc** later.

There is in fact another perspective for ∞ -categories, where we do not get in deep here besides the statements of the major results.

Definition 5.10. *An **n -type** is a CW complex X such that $\pi_i(X, x_0) = 0$ for every $i > n$ and $x_0 \in X$.*

By observation, we see that

Proposition 5.11. *A 0-type is essentially a set, and a 1-type is essentially a groupoid. The equivalence between the categories here are denoted by $\prod_0$ and $\prod_1$.*

In fact, we have the similar equivalence for larger n :

Theorem 5.12. *We have the following equivalences:*

- (i) $\prod_n : \mathbf{nType} \rightarrow \mathbf{nGpd}$ as $(n+1)$ -categories, for every $n \geq 0$.
- (ii) $\prod_\infty : \infty\mathbf{Type} \rightarrow \infty\mathbf{Gpd}$ as $(\infty, 1)$ -categories.
- (iii) $\prod_\infty : \mathbf{CW} \rightarrow \mathbf{Kan}$ as $(\infty, 1)$ -categories.

More discussion about these can be found in [Lur09].

Simplicial Category

Definition 5.13. *A **simplicial category** is defined as a **sSet**-enriched category.*

Explicitly, if $\mathcal{C}$ is a simplicial category, then $\mathcal{C}(A, B)$ is a simplicial set for every A and B in $\mathcal{C}$. There also exists composition:

$$\mathcal{C}(A, B) \times \mathcal{C}(B, C) \rightarrow \mathcal{C}(A, C)$$

*which is a morphism in **sSet**. Identity and associativity is defined similarly as any enriched category.*

Remark 20. *We have a natural functor $F : \mathbf{sSet} \rightarrow \mathbf{Set}$ by sending X to X_0 . F therefore defines a functor from **sSet** – **Cat** to **Cat**.*

Definition 5.14. *Let $F : \mathcal{V} \rightarrow \mathbf{Set}$ be a functor. Given $\mathcal{C}$ a $\mathcal{V}$ -enriched category, we define $\mathcal{C}(A, B)$ to be $F(\mathcal{C}(A, B)) \in \mathbf{Set}$ and $\underline{\mathcal{C}}(A, B)$ to be the original $\mathcal{C}(A, B) \in \mathcal{V}$.*

We wish to show that **sSet** has a simplicial enrichment; namely, we define $\underline{\mathbf{sSet}}(X, Y) = Y^X$.

Notice that by 3.8, the exponential object exists for **sSet**, so our definition is valid. By the definition of Y^X provided in that theorem, we have that $Y_n^X = \mathbf{sSet}(X \times \Delta^n, Y)$.

Remark 21. *Notice that in any Cartesian closed category $\mathcal{C}$, we could define a $\mathcal{C}$ -enrichment for $\mathcal{C}$. Namely, we define $\underline{\mathcal{C}}(X, Y) = Y^X$.*

To define composition $Y^X \times Z^Y \rightarrow Z^X$, we first define a morphism in the following diagram:

$$\begin{array}{ccc} & & Y \times Z^Y \\ & \nearrow^{ev \times 1_{Z^Y}} & \searrow_{ev} \\ X \times Y^X \times Z^Y & \dashrightarrow & Z \end{array}$$

and use the one-to-one correspondence between $\mathcal{C}(X \times Y^X \times Z^Y, Z)$ and $\mathcal{C}(Y^X \times Z^Y, Z^X)$.

Definition 5.15. Given X, Y in **Top**, $C(X, Y)$ can be seen as a topological space with respect to compact open topology. However, it is not an exponential object unless X is locally compact Hausdorff.

Therefore, we now restrict **Top** to locally compact Hausdorff spaces, and **Top** has exponential objects now, and therefore **Top** has a **Top**-enrichment, by defining $\mathbf{Top}(X, Y) = Y^X = C(X, Y)$.

Notice that we have $S : \mathbf{Top} \rightarrow \mathbf{sSet}$, so **Top** now has a **sSet**-enrichment (in fact a **Kan**-enrichment by 3.5). Namely,

$$\underline{\mathbf{Top}}(X, Y) = SC(X, Y).$$

In fact, now only does **Top** has a **Kan**-enrichment but also does **Kan** itself.

Proposition 5.16. If $X \in \mathbf{sSet}$ and $Y \in \mathbf{Kan}$, then so is Y^X .

Proof. The proof uses the adjunction between $\mathbf{sSet}(-, Y^X)$ and $\mathbf{sSet}(- \times X, Y)$. We therefore can turn the diagram on the left to the one on the right, where the existence of horn fillings is clear.

$$\begin{array}{ccc} \Lambda_k^n & \longrightarrow & Y^X \\ \downarrow & \nearrow \text{dashed} & \\ \Delta^n & & \end{array} \quad \begin{array}{ccc} X \times \Lambda_k^n & \longrightarrow & Y \\ \downarrow & \nearrow \text{dashed} & \\ X \times \Delta^n & & \end{array}$$

□

Homotopy Coherent Nerve

We often use **sCat** to denote the category of simplicial categories.

Definition 5.17. We now try to define the *homotopy coherent nerve*, a functor that sends a simplicial category to a simplicial set.

As usual, we define $(NC)_n = \mathbf{sCat}(F[n], \mathcal{C})$, where $F : \Delta \rightarrow \mathbf{sCat}$ is a functor sending simplices to simplicial categories.

The definition for the functor F is more involved, and it is put into the Appendix.

By the construction, we know that NC is a simplicial set, and it can be shown that $(NC)_0$ corresponds to objects in $\mathcal{C}$; in fact, we have that

Theorem 5.18. NC is a quasi-category if $\mathcal{C}$ is **Kan**-enriched.

A proof for this can be found in [Lur09].

Now we have the other definition for the infinity category of spaces, **Spc**.

The ∞ -category of Spaces

Definition 5.19. $\mathbf{Spc} = N(\mathbf{Kan})$, where **Kan** is viewed as a **Kan**-enriched category (therefore a simplicial category).

To define the notion of equivalence between quasi-categories, we need the following definitions:

Definition 5.20. The *mapping space* in a simplicial set $\mathcal{C}$ between two objects $X, Y \in \mathcal{C}_0$ is defined to be the simplicial set in the pullback below:

$$\begin{array}{ccc} \mathcal{C}(X, Y) & \longrightarrow & \mathcal{C}^{\Delta^1} \\ \downarrow & & \downarrow \\ \Delta^0 & \xrightarrow{\langle X, Y \rangle} & \mathcal{C}^{\Delta^0 + \Delta^0} \end{array}$$

Intuitively, we view every morphism $X \rightarrow Y$ in $\mathcal{C}$ to be embedded in all the morphisms in $\mathcal{C}$, represented by $\mathcal{C}^{\Delta^1}$. The map $\mathcal{C}^{\Delta^1} \rightarrow \mathcal{C}^{\Delta^0 + \Delta^0}$ is induced by the boundary maps from Δ^1 to Δ^0 .

The definition gives the set of morphisms between any two objects a simplicial structure, and, in fact, there's more to say for a quasi-category.

Theorem 5.21. $\mathcal{C}(X, Y)$ is a Kan complex if $\mathcal{C}$ is a quasi-category.

A proof for this can be found in [Lur09].

A simplicial map between two quasi-categories is going to be called a functor, and an equivalence is defined based on the following notions:

Definition 5.22. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ in **qCat** is

- (i) **essentially surjective** if for any $Y \in \mathcal{D}$, there exists some $X \in \mathcal{C}$ such that $FX \simeq Y$ in $\mathcal{D}$ (i.e., FX and Y are connected by a homotopy equivalence);
- (ii) **fully faithful** if $F : \mathcal{C}(X, Y) \rightarrow \mathcal{D}(FX, FY)$ is a homotopy equivalence for every $X, Y \in \mathcal{C}$ (i.e., the two simplicial sets are homotopy equivalent after geometric realization);
- (iii) an **equivalence** if F is fully faithful and essentially surjective.

Theorem 5.23. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ induces morphisms on each of the mapping space: $F : \mathcal{C}(X, Y) \rightarrow \mathcal{D}(FX, FY)$.

Now, viewing **Top** as a simplicial category, we have an equivalence:

Theorem 5.24. $N(\mathbf{Top}) \simeq \mathbf{Spc}$.

Theorem 5.25. Given a **Kan**-enriched category $\mathcal{C}$, we have the following commutative diagram (up to equivalence):

$$\begin{array}{ccc} & & \mathbf{HoC} \\ & \nearrow & \downarrow \simeq \\ \mathcal{C} & \longrightarrow NC & \longrightarrow \mathbf{HoNC} \end{array}$$

where $\mathcal{C} \rightarrow NC$ sends zero-simplices to objects in NC and the Kan complex $\mathcal{C}(X, Y)$ to the mapping space $NC(X, Y)$.

The idea of this diagram is that we sends an object $X \in \mathcal{C}$ to the object in NC that has all simplices a constant chain $X \rightarrow X \rightarrow \dots \rightarrow X$ filled by identities. This gives a natural embedding from $\mathcal{C}$ to NC , where the embedding for the mapping spaces is induced according to 5.23.

Theorem 5.26. In fact, we have the following diagram which commutes up to an equivalence:

$$\begin{array}{ccccc} & & \mathbf{Kan} & & \\ & \nearrow S & \downarrow & \searrow & \\ \mathbf{Top} & \xrightarrow{\Pi_\infty} & \mathbf{Spc} & \longrightarrow & \mathbf{HoKan} \\ & \searrow & \downarrow & \swarrow | - | & \\ & & \mathbf{HoTop} & & \end{array}$$

where Π_∞ is the functor sending each mapping space in **Top** to the corresponding ∞ -groupoid.

The bottom-left and top-right triangles commute according to the theorem above. The bottom-right one commute as the equivalence between $N\mathbf{Kan}$ and $N\mathbf{Top}$ is induced by the adjoint functors S and $| - |$, so taking the homotopy category of their nerves we get this triangle. The top-left one requires some extra universal property about **Kan**.

Remark 22. *Something that makes the result non-trivial is the fact that $\mathbf{Spc} \not\cong \mathbf{HoKan}$. This can be seen from the use of mapping space.*

In $\mathbf{HoKan}$, all mapping spaces are contractible. Y^X is itself a Kan complex, and therefore for any simplicial map $\Delta^n \rightarrow Y^X$, we can first restrict it to $\Delta_k^n \rightarrow Y^X$ for some k (which is doable as the horn is a deformation retract of the simplex), and then induct on n by viewing the horn as the colimit of a number of $n - 1$ -simplices.

On the other hand, it can be shown that mapping spaces in $\mathbf{Spc}$ are generally not contractible. According to the embedding $\mathcal{C} \rightarrow NC$ as depicted above, we have that $X^A = NC(A, X) \simeq \mathcal{C}(A, X)$, where on the right hand side A and X are objects in $\mathcal{C}$ and on the left hand side are their embeddings into NC . In particular, if $A = 1$, the terminal object, then $\mathcal{C}(1, X) \cong X$ which is not contractible if X is not.

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